

WTW 158 Calculus
ASSIGNMENT: Exam July 2021

When submitting this assignment, you agree to the following:

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, assignments, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, assessments, assignments, examinations and/or any other forms of assessment.

This assignment consists of 2 parts.

Section B: 15 marks (minimum 30 minutes)

Section C: 15 marks (35 minutes)

INSTRUCTIONS:

1. The exam paper is an assignment consisting of 2 parts.
2. Draw a line with a ruler at the end of each question.
3. Write your surname, initials and student number clearly in the beginning of your assignment.
4. **Show all your steps and calculations.**
5. Only handwritten assignments submitted in one PDF file will be accepted.
6. Before submitting, make sure that you are satisfied with your answers and that you submit the correct document.
7. No late assignments will be accepted.
8. No assignments sent via e-mail will be accepted.
9. Calculators **may not** be used.

Section B.

Question 1 [6 marks]

- (1.1) Suppose f is a differentiable function. Use the definition of a derivative to find the derivative of (2)

$$g(x) = xf(x).$$

- (1.2) Consider function f given below and determine where the slope of the function is 1, if possible. (2)

$$f(x) = e^{\sqrt{x}} + x.$$

- (1.3) Determine the interval(s) on which the function f is increasing, where (2)

$$f(x) = \ln(4 - x^2).$$

Question 2 [5 marks]

- (2.1) For $a, b, c, d > 0$, evaluate the limit (if it exists): (2)

$$\lim_{x \rightarrow 0} \frac{a^x - b^x}{c^x - d^x}.$$

- (2.2) Find the horizontal asymptotes of the function: (3)

$$f(x) = \frac{\sqrt{x^2 + 1}}{3x - 5}.$$

Question 3 [4 marks]

Consider the points $A(6, 0, k)$ and $B(1, k, -3)$ in the three dimensional space.

- (3.1) Determine the value(s) of k , such that $\overline{OA}$ and $\overline{OB}$ are orthogonal to each other. (2)

- (3.2) When adding vectors $\overline{OA}$ and $\overline{AB}$, we place the starting point of vector $\overline{AB}$ at the endpoint of vector $\overline{OA}$ and the sum of these two vectors then is the resultant vector $\overline{OB}$. Let $k = 1$ and determine vector $\overline{AB}$ and the length of vector $\overline{AB}$. (2)